

# Mobile Testing

Android, iOS, real devices, emulators, Appium and device-cloud platforms.

## What changes in Mobile Testing

Mobile applications deal with different operating systems, screen sizes, hardware, permissions, networks and interruptions. Testing only the functional flow is not enough.

## Native, hybrid, WebView and PWA

Native apps use platform-specific technologies. Hybrid apps combine web technologies with a native container. WebView embeds web content. A PWA is a web application with progressive capabilities.

## Real device vs emulator/simulator

Emulators and simulators are useful for speed and initial coverage. Real devices are important for hardware, performance, camera, biometrics, notifications and real network behavior.

## Essential scenarios

Installation, upgrade, login, permissions, rotation, background/foreground, calls, battery, GPS, camera, keyboard, offline/online, slow networks and interruptions should be considered according to the product.

## Appium

Appium is a mobile automation framework that drives applications through platform automation APIs. It can be combined with languages such as Python, Java and JavaScript.

## BrowserStack and device clouds

Device-cloud platforms provide remote access to browsers and real devices. They do not replace Appium; they can host automated and manual executions across many combinations.

## Android and iOS

Android uses artifacts such as APK/AAB and tools such as ADB/Android Studio. iOS uses IPA and the Xcode ecosystem. Restrictions, permissions and distribution differ.

## Mobile strategy

Prioritize device combinations using user data, analytics, risk and official support. Infinite coverage is impossible; build a defensible test matrix.